



1. One day after undergoing uncomplicated splenectomy for management of injuries sustained in a motor vehicle collision, a 43-year-old man has the sudden onset of shortness of breath while receiving a transfusion of packed red blood cells. He also has moderate pain at the intravenous site and moderate back pain. He has no other history of serious illness. His medications are hydromorphone and subcutaneous heparin. He is diaphoretic. His temperature is 37.2°C (99°F), pulse is 90/min, respirations are 24/min, and blood pressure is 120/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Examination of the skin shows no abnormalities. Breath sounds are clear bilaterally; no wheezes or rhonchi are heard. Abdominal examination shows a well-healing surgical incision. Which of the following is the most likely mechanism of this patient's condition?
- ☐ A) Endotoxin release from infused bacteria
 - ☐ B) Immune response to transfused plasma proteins
 - ☐ C) Intra-abdominal hemorrhage
 - ☐ D) Preformed antibodies to transfused red blood cells



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2. A 62-year-old woman has shortness of breath immediately after beginning transfusion of fresh frozen plasma and 2 units of packed red blood cells to manage acute lower gastrointestinal bleeding. She has been hospitalized for 24 hours. Five years ago, she underwent mechanical valve replacement for treatment of aortic stenosis. She has no other history of serious illness. Her only outpatient medications are warfarin and lisinopril, which have been discontinued since admission. Her temperature is 37.8°C (100°F), pulse is 110/min and regular, respirations are 28/min, and blood pressure is 100/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 82%. Crackles are heard from the lung bases to the midlung fields. There is cyanosis of the upper and lower extremities. A urinary catheter is draining clear urine. A chest x-ray is pending. Which of the following is the most likely diagnosis?

- ☐ A) Acute hemolytic transfusion reaction
- ☐ B) Pneumonia
- ☐ C) Pulmonary embolism
- ☐ D) Sepsis
- ☐ E) Transfusion-related acute lung injury

3. An 18-year-old man is brought to the emergency department 30 minutes after sustaining injuries to the head, chest, and abdomen when he was thrown from his motorcycle. En route to the hospital, he received 2 L of crystalloid. On arrival, his temperature is 35.9°C (96.6°F), pulse is 110/min, and blood pressure is 115/90 mm Hg. Glasgow Coma score is 11. Examination shows multiple ecchymoses and abrasions. His hematocrit is 29%. Two units of packed red blood cells are administered. A CT scan of the head shows diffuse axonal injuries, and an x-ray of the chest shows a pulmonary contusion without pneumothorax. A CT scan of the abdomen is shown. Which of the following is the most appropriate next step in management of his abdominal injuries?

- ☐ A) Serial examinations in the intensive care unit
- ☐ B) Angiography
- ☐ C) Ultrasonography
- ☐ D) Repeat CT scan in 24 hours
- ☐ E) Diagnostic peritoneal lavage
- ☐ F) Laparotomy





4. A 20-year-old man is brought to the emergency department 50 minutes after being involved in a motor vehicle collision. He was an unrestrained passenger. On arrival, he is alert. He says he has neck pain and numbness in both legs; he cannot move his legs. His temperature is 37.2°C (99°F), pulse is 70/min, respirations are 20/min and unlabored, and blood pressure is 80/50 mm Hg. Examination shows a patent airway. There is no jugular venous distention. There are multiple contusions over the trunk and a bony deformity of the right calf. The cervical spine is tender. Sensation to touch is absent below the midchest. Abdominal ultrasonography for trauma (FAST) shows no free fluid in the pelvis. X-rays of the cervical spine show subluxation of C7 on T1. X-rays of the chest and pelvis show no abnormalities. An x-ray of the right lower extremity shows comminuted fractures of the tibia and fibula. CT scans of the abdomen and pelvis show a small contained hematoma. After infusion of 2 L of crystalloid solution, the patient's blood pressure does not improve. Which of the following is the most likely cause of this patient's persistent hypotension?

- ☐ A) Aortic disruption
- ☐ B) Bleeding from the fractures
- ☐ C) Closed head injury
- ☐ D) Hepatic injury
- ☐ E) Spinal cord injury



5. An 18-year-old woman comes to the emergency department because of a 3-week history of headache. The headache has worsened today and is now accompanied by mildly blurred vision. Her mother has a history of migraines. The patient is 163 cm (5 ft 4 in) tall and weighs 91 kg (200 lb); BMI is 34 kg/m². Her temperature is 37°C (98.6°F), pulse is 76/min, respirations are 16/min, and blood pressure is 140/80 mm Hg. Funduscopic examination shows bilateral papilledema. The remainder of the examination shows no abnormalities. A CT scan of the head without contrast shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Optical coherence tomography
 - ☐ B) Intravenous administration of methylprednisolone
 - ☐ C) Intravenous administration of nifedipine
 - ☐ D) Visual evoked potentials
 - ☐ E) Lumbar puncture

6. A 76-year-old man comes to the physician because of a lesion on his nose that has been gradually increasing in size during the past 5 years. He has hypercholesterolemia treated with atorvastatin. Examination shows no lymphadenopathy. There is a 1-cm lesion on the nose. A photograph is shown. Which of the following is the most likely diagnosis?

- ☐ A) Actinic keratosis
- ☐ B) Basal cell carcinoma
- ☐ C) Hemangioma
- ☐ D) Melanoma
- ☐ E) Squamous cell carcinoma





7. A 65-year-old man comes to the office because of a 6-month history of an enlarging lesion on his left forearm. The lesion is not painful or itchy; it does not bleed and has not changed shape. He has no history of serious illness and takes no medications. Vital signs are within normal limits. A photograph of the lesion is shown. Which of the following is the most appropriate next step in this patient's management?
- ☐ A) Excision with 1-cm margins
 - ☐ B) Excision with 2-cm margins
 - ☐ C) Excision with 5-cm margins
 - ☐ D) Sentinel node biopsy and excision with 1-cm margins
 - ☐ E) Sentinel node biopsy and excision with 2-cm margins



7. A 65-year-old man comes to the office because of a 6-month history of an enlarging lesion on his left forearm. The lesion is not painful or itchy; it does not bleed and has not changed shape. He has no history of serious illness and takes no medications. Vital signs are within normal limits. A photograph of the lesion is shown. Which of the following is the most appropriate next step in this patient's management?
- ☐ A) Excision with 1-cm margins
 - ☐ B) Excision with 2-cm margins
 - ☐ C) Excision with 5-cm margins
 - ☐ D) Sentinel node biopsy and excision with 1-cm margins
 - ☐ E) Sentinel node biopsy and excision with 2-cm margins
 - ☐ F) Sentinel node biopsy and excision with 5-cm margins



8. A 47-year-old man is brought to the emergency department by paramedics after rescue workers found him on a snow-covered mountain. At the scene, he was alert and oriented but shivering and unable to walk without assistance. He said he had become lost while cross-country skiing. On arrival, he is shivering and reports that his toes are numb. His temperature is 36°C (96.8°F), pulse is 60/min, respirations are 16/min, and blood pressure is 120/65 mm Hg. Examination shows pale toes bilaterally. Blisters are noted on the 4th and 5th toes bilaterally. Sensation to touch is absent over the toes bilaterally. Pedal pulses are faintly palpable in both feet. Ankle brachial indices are 0.85 bilaterally (N>1). Tetanus toxoid is administered, and the feet are immersed in a 104°F water bath for 20 minutes. During immersion, his feet become painful. In addition to administration of acetaminophen, which of the following is the most appropriate next step in management?

- ☐ A) Hospital admission for observation
- ☐ B) Application of nitroglycerin ointment to the feet
- ☐ C) Intravenous infusion of unfractionated heparin
- ☐ D) Immediate arteriography of the lower extremities
- ☐ E) Amputation of all toes with blisters at 24 hours post injury



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9. An 87-year-old man is transferred to the hospital from a skilled nursing care facility 6 hours after staff noticed black skin on his right leg. He has had three strokes during the past 2 years. He has hypertension and type 2 diabetes mellitus. Current medications include metoprolol, glipizide, and clopidogrel. He does not smoke. He is not communicative and responds only to painful stimuli. There are severe contractures of the lower extremities. Examination of the lower extremities shows black skin over the right ankle and foot. Pulses are absent in the lower extremities distally. Extension of the hips is limited to 110 degrees; extension of the knees is limited to 70 degrees. Which of the following is the most appropriate next step in management?

- ☐ A) Cilostazol therapy
- ☐ B) Arteriography with contrast
- ☐ C) Percutaneous balloon angioplasty
- ☐ D) Surgical revascularization
- ☐ E) Amputation



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10. A 47-year-old woman comes to the physician because of a 2-year history of a lump on her right ankle that seems to have increased in size during the past 2 months. The lump is slightly painful only when she is active. She has had no other symptoms. She has no history of serious illness and takes no medications. Examination of the right ankle shows a 2.8-cm raised lesion that is firm and adherent to the bone. There are no skin changes or lymphadenopathy. Range of motion of the ankle is full. X-rays of the right lower extremity are shown. Which of the following is the most appropriate next step in management?

☐ A) Nonweight-bearing exercises



10. A 47-year-old woman comes to the physician because of a 2-year history of a lump on her right ankle that seems to have increased in size during the past 2 months. The lump is slightly painful only when she is active. She has had no other symptoms. She has no history of serious illness and takes no medications. Examination of the right ankle shows a 2.8-cm raised lesion that is firm and adherent to the bone. There are no skin changes or lymphadenopathy. Range of motion of the ankle is full. X-rays of the right lower extremity are shown. Which of the following is the most appropriate next step in management?

- ☐ A) Nonweight-bearing exercises
- ☐ B) Second x-ray in 2 months
- ☐ C) Bone scan
- ☐ D) DEXA scan
- ☐ E) Biopsy of the lesion



11. Three days after undergoing emergency partial gastrectomy for upper gastrointestinal hemorrhage, a 72-year-old man has increasing abdominal pain. His postoperative course had been uncomplicated. He has hypertension treated with lisinopril and type 2 diabetes mellitus treated with glipizide. He underwent coronary angioplasty with stent placement 2 years ago. His current medications are patient-controlled morphine and subcutaneous unfractionated heparin. He is anxious and diaphoretic. His temperature is 37.2°C (99°F), pulse is 110/min, respirations are 20/min, and blood pressure is 145/90 mm Hg. Examination of the abdomen shows an upper midline incision without signs of infection. The remainder of the examination shows no other abnormalities. His hematocrit is 32%, and leukocyte count is 8000/mm³. In addition to administering morphine, which of the following is the most appropriate next step in management?

- ☐ A) Serum D-dimer assay
- ☐ B) 12-Lead ECG
- ☐ C) CT scan of the abdomen
- ☐ D) Upper endoscopy
- ☐ E) Local wound exploration at bedside

12. A 62-year-old woman comes to the physician for a routine health maintenance examination. She does not smoke and only drinks socially. A systolic ejection murmur is best heard over the second right intercostal space with transmission to the carotid arteries. An ECG shows normal sinus rhythm and a pattern of left ventricular hypertrophy with strain. Which of the following manifestations would indicate the need for aortic valve replacement?
- ☐ A) Angina and syncope
 - ☐ B) Generalized weakness and fatigue
 - ☐ C) Peripheral edema and ascites
 - ☐ D) Pulmonary embolus
 - ☐ E) Ventricular arrhythmia

13. A 77-year-old man comes to the emergency department 5 hours after he was awakened by severe pain in his left leg. He reports that his left foot feels numb and he cannot move his foot and toes. He has a 4-year history of stable claudication of both calves, a 40-year history of hypercholesterolemia, and a 15-year history of type 2 diabetes mellitus. He was recently diagnosed with atrial fibrillation. Medications include aspirin and digoxin daily. He is in mild distress. His pulse is 84/min and irregularly irregular, and blood pressure is 145/85 mm Hg. The left lower extremity is pale and cool below the knee. Femoral pulses are 1+ on the right and absent on the left. Which of the following is the most likely diagnosis?
- ☐ A) Compartment syndrome
 - ☐ B) Deep venous thrombosis
 - ☐ C) Left femoral artery embolism
 - ☐ D) Rest pain secondary to progressive atherosclerotic peripheral vascular disease
 - ☐ E) Thrombosis of the left superficial femoral artery

14. A 67-year-old woman develops swelling of her left leg 3 days after Roux-en-Y choledochojejunostomy for an obstructing cancer of the head of the pancreas. Venous Doppler ultrasonography is positive. Which of the following is the most appropriate next step in management?
- ☐ A) Elevation of the left lower extremity
 - ☐ B) Use of compression stockings
 - ☐ C) Physical therapy
 - ☐ D) Intravenous streptokinase therapy
 - ☐ E) Anticoagulant therapy with heparin

15. Two hours after admission to the hospital for injuries sustained in a motor vehicle collision, a 77-year-old woman has the onset of tachycardia. She was the restrained driver of a vehicle that struck a tree head on. She sustained bilateral anterior fractures of the 5th and 6th ribs. On admission, ECG showed occasional premature atrial contractions, and 1 L of lactated Ringer solution was administered. Vital signs on admission and now are:

	Two Hours Ago on Admission	Now
Temperature	37.0°C (98.6°F)	37.4°C (99.3°F)
Pulse	86/min	100/min and irregular
Respirations	20/min	24/min
Blood pressure	110/60 mm Hg	110/60 mm Hg

Examination shows no jugular venous distention or pulsus paradoxus. Posterior crackles are heard over the lower quarter of the lung fields bilaterally. Hematocrit is 43%. Which of the following is the most likely cause of the cardiovascular findings?

- ☐ A) Blunt cardiac injury
- ☐ B) Cardiomyopathy
- ☐ C) Cor pulmonale
- ☐ D) Hypovolemic shock
- ☐ E) Pericardial tamponade



16. A 37-year-old man comes to the physician with his wife because of a 10-week history of severe shortness of breath after walking one block and swelling of his legs. His wife says that he has snored for many years and during the past 6 months has had many episodes every night in which he stops breathing for 15 seconds and then suddenly wakes up. He has had constant daytime fatigue for the past 3 months. He has type 2 diabetes mellitus and hypertension. Current medications include insulin and an ACE inhibitor. He appears cyanotic and is in mild respiratory distress. He is 173 cm (5 ft 8 in) tall and weighs 145 kg (320 lb); BMI is 49 kg/m². His pulse is 90/min, respirations are 24/min, and blood pressure is 144/88 mm Hg. Examination shows decreased air movement at the lung bases. Scattered rhonchi are heard. Heart sounds are distant. Examination of the lower extremities shows brawny induration and 3-cm granulating ulcers above the medial malleoli. There is 3+ pitting edema to the midtibia bilaterally. Arterial blood gas analysis on room air shows a Pco₂ of 50 mm Hg and a Po₂ of 52 mm Hg. If this patient is not treated for the current findings, he will be at increased risk for which of the following?

- ☐ A) Aortic dissection
- ☐ B) Cor pulmonale
- ☐ C) Intracerebral hemorrhage
- ☐ D) Iron deficiency anemia
- ☐ E) Thrombotic stroke



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17. A 62-year-old man is brought to the emergency department 3 hours after the onset of shortness of breath at rest. One week ago, he was discharged from the hospital after undergoing a partial colectomy for a diverticular abscess. His postoperative course had been uncomplicated. On arrival, he is in mild respiratory distress. His temperature is 37°C (98.6°F), pulse is 110/min, respirations are 22/min, and blood pressure is 140/98 mm Hg. The lungs are clear to auscultation. Abdominal examination shows a well-healing incision with no erythema or tenderness. Pulse oximetry on room air shows an oxygen saturation of 89%. Oxygen is administered via nasal cannula, and his symptoms improve. Laboratory studies are pending. An x-ray of the chest shows no abnormalities. An ECG shows sinus tachycardia. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance
- ☐ B) Aspirin therapy
- ☐ C) Heparin therapy
- ☐ D) Metronidazole therapy
- ☐ E) Transfusion of 2 units of packed red blood cells



18. One hour after undergoing a colon resection for a sigmoid carcinoma, a 57-year-old woman is tachypneic and agitated in the recovery room. She was extubated in the operating room and is breathing 30% oxygen by nasal cannula. She is 178 cm (5 ft 10 in) tall and weighs 140 kg (308 lb); BMI is 44 kg/m². Her pulse is 120/min, respirations are 40/min, and blood pressure is 135/90 mm Hg. Breath sounds are shallow and symmetrical. Heart sounds are normal. Arterial blood gas analysis shows:

pH	7.25
PCO ₂	60 mm Hg
PO ₂	70 mm Hg

Which of the following is the most appropriate next step in management?

- ☐ A) Administration of a fluid bolus
- ☐ B) Administration of naloxone
- ☐ C) Diuretic therapy
- ☐ D) Fiberoptic bronchoscopy
- ☐ E) Heparin therapy
- ☐ F) Intubation
- ☐ G) Laryngoscopy



19. A 42-year-old man comes to the office for examination prior to undergoing gastric bypass. He reports a 2-month history of shortness of breath after walking less than one-half block. He has hypertension treated with hydrochlorothiazide. He is 180 cm (5 ft 11 in) tall and weighs 171 kg (376 lb); BMI is 52 kg/m². His pulse is 84/min, respirations are 14/min, and blood pressure is 132/82 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Decreased breath sounds are heard bilaterally. Decreased heart sounds are heard. Arterial blood gas analysis on room air shows:

pH	7.31
Pco ₂	56 mm Hg
PO ₂	72 mm Hg

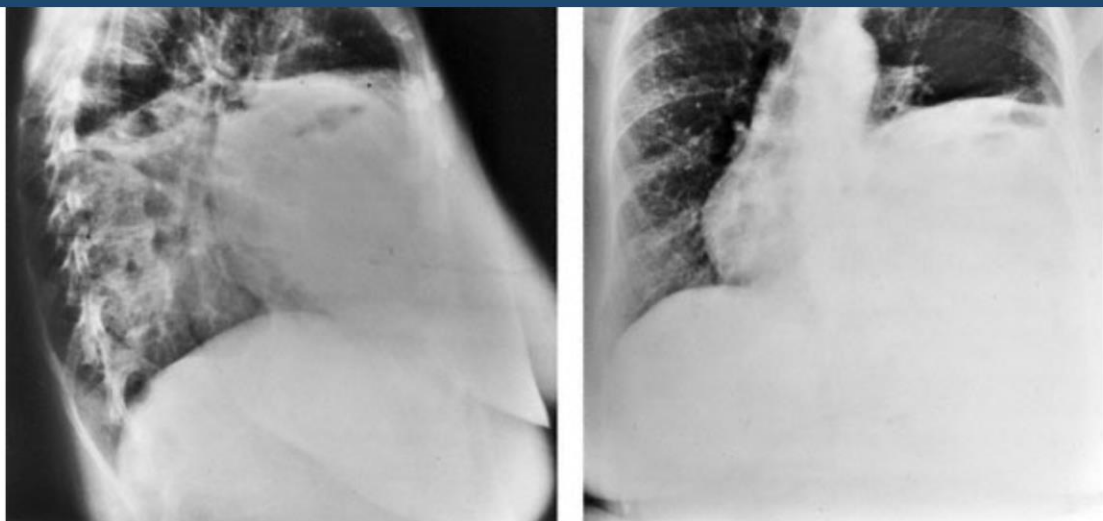
Which of the following mechanisms is the most likely cause of these findings?

- ☐ A) Decreased diffusion capacity
- ☐ B) Decreased ventilation
- ☐ C) Increased diffusion capacity
- ☐ D) Increased ventilation
- ☐ E) Left-to-right shunting
- ☐ F) Right-to-left shunting
- ☐ G) Ventilation-perfusion mismatch





20. A 32-year-old woman is brought to the emergency department 30 minutes after being involved in a head-on motor vehicle collision. She was the unrestrained front seat passenger. She is alert. Her temperature is 36.8°C (98.2°F), pulse is 110/min, respirations are 16/min, and blood pressure is 140/85 mm Hg. Examination shows ecchymoses over the forehead, left side of the chest, and upper abdomen. Pupils are equal and reactive to light. There is no jugular venous distention. Breath sounds are decreased over the left lung base. Heart sounds are normal. Abdominal examination shows epigastric tenderness. X-rays of the chest are shown. Which of the following is the most likely diagnosis?
- ☐ A) Aortic rupture
 - ☐ B) Diaphragmatic rupture
 - ☐ C) Flail chest



20. A 32-year-old woman is brought to the emergency department 30 minutes after being involved in a head-on motor vehicle collision. She was the unrestrained front seat passenger. She is alert. Her temperature is 36.8°C (98.2°F), pulse is 110/min, respirations are 16/min, and blood pressure is 140/85 mm Hg. Examination shows ecchymoses over the forehead, left side of the chest, and upper abdomen. Pupils are equal and reactive to light. There is no jugular venous distention. Breath sounds are decreased over the left lung base. Heart sounds are normal. Abdominal examination shows epigastric tenderness. X-rays of the chest are shown. Which of the following is the most likely diagnosis?
- ☐ A) Aortic rupture
 - ☐ B) Diaphragmatic rupture
 - ☐ C) Flail chest
 - ☐ D) Hemothorax
 - ☐ E) Pneumothorax
 - ☐ F) Pulmonary contusion
 - ☐ G) Tracheal laceration



21. Seven days after undergoing a splenectomy for refractory bleeding after being kicked by a horse, a 25-year-old woman has a temperature of 39°C (102.2°F) and malaise. She received 8 units of packed red blood cells during the first 24 hours after the operation. Her postoperative course had been uncomplicated until today. Her pulse is 110/min, and blood pressure is 120/70 mm Hg. Examination shows a well-healing surgical incision without erythema or tenderness. Breath sounds are decreased on the left. The abdomen is soft with mild fullness in the left upper quadrant. Her hematocrit is 30%, leukocyte count is 25,000/mm³ with a left shift, and platelet count is 800,000/mm³. An x-ray of the chest shows a left pleural effusion. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Ultrasonography of the gallbladder
- ☐ B) Upper gastrointestinal series
- ☐ C) CT scan of the abdomen
- ☐ D) Liver-spleen scan
- ☐ E) Bronchoscopy



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22. A previously healthy 25-year-old woman comes to the office because of a 4-month history of intermittent abdominal cramps. During this period, she has noticed occasional mucus mixed with her stools but has never noted any bright red blood. She is 168 cm (5 ft 6 in) tall and weighs 45 kg (100 lb); BMI is 16 kg/m². Her vital signs are within normal limits. Abdominal examination shows mild tenderness to palpation of the right lower quadrant; there is no distention, guarding, or rebound. Rectal examination shows normal sphincter tone, a noninflamed anal fistula, brown stool, and no masses. Test of the stool for occult blood is positive. Which of the following is the most likely diagnosis?

- ☐ A) Crohn disease
- ☐ B) Diverticulitis
- ☐ C) Endometriosis
- ☐ D) Gastritis
- ☐ E) Irritable bowel syndrome
- ☐ F) Ischemic colitis
- ☐ G) Ulcerative colitis



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23. A 16-year-old boy is brought to the emergency department because of a 6-month history of intermittent bright red rectal bleeding. He has no history of serious illness and takes no medications. Vital signs are within normal limits. Cardiopulmonary examination shows no abnormalities. Digital rectal examination shows blood in the rectal vault. Endoscopy shows scattered 1-cm to 2-cm polyps throughout the colon. Endoscopic polypectomy of the larger polyps is performed. Results of biopsy of the polyps show dilated, cystic, mucus-filled glands with abundant lamina propria and inflammatory infiltrates. Which of the following is the most likely diagnosis?

- ☐ A) Cowden disease
- ☐ B) Cronkite-Canada syndrome
- ☐ C) Familial adenomatous polyposis
- ☐ D) Hereditary nonpolyposis colorectal cancer
- ☐ E) Juvenile polyposis
- ☐ F) Peutz-Jeghers syndrome

24. A 33-year-old man comes to the office as a new patient after he recently moved to the state. His last visit to a physician was 2 years ago. He has a 13-year history of ulcerative colitis well controlled with mesalamine. His mother has type 2 diabetes mellitus. His father was diagnosed with colon cancer at the age of 55 years. The patient does not smoke cigarettes. He drinks two 12-oz beers monthly. Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate colon cancer screening for this patient?

- ☐ A) Colonoscopy now
- ☐ B) Colonoscopy at the age of 45 years
- ☐ C) Colonoscopy at the age of 50 years
- ☐ D) Virtual colonoscopy now
- ☐ E) Virtual colonoscopy at the age of 45 years
- ☐ F) Virtual colonoscopy at the age of 50 years

25. A 67-year-old man with a 2-year history of gastroesophageal reflux disease comes to the physician for a follow-up examination. Treatment with the maximum dosage of proton pump inhibitors has not relieved his symptoms. He also has hypertension and hypercholesterolemia. His only other medication is hydrochlorothiazide. His blood pressure is 135/85 mm Hg. Examination shows no other abnormalities. Esophagogastroduodenoscopy (EGD) shows grade III esophagitis and Barrett esophagus. A biopsy specimen of the esophagus shows mild dysplasia. He undergoes Nissen fundoplication, and his symptoms resolve. Which of the following is the most appropriate next step in management?
- ☐ A) Continue proton pump inhibitor therapy
 - ☐ B) Repeat the EGD if the patient's symptoms recur
 - ☐ C) Repeat the EGD and biopsy in 6 months
 - ☐ D) Perform laser ablation of Barrett esophagus
 - ☐ E) No further treatment is indicated

26. A 67-year-old woman is admitted to the hospital through the emergency department for evaluation of rectal bleeding that began 6 hours ago. Results of colonoscopy 2 years ago showed several scattered diverticula. On admission, vital signs are within normal limits. The abdomen is nontender, and she has not had any abdominal pain. Digital rectal examination discloses gross blood. Results of laboratory studies show:

Hemoglobin	6.8 g/dL
Leukocyte count	8000/mm ³
Platelet count	215,000/mm ³

Colonoscopy is done and shows maroon blood and stool throughout the colon. Radiolabeled erythrocyte scan shows activity in the right lower quadrant. Visceral angiography does not demonstrate active contrast extravasation. During the procedure, the patient receives a transfusion of 3 units of packed red blood cells; repeat hemoglobin concentration obtained after transfusion is 6.6 g/dL. Which of the following is the most appropriate next step in management?

- ☐ A) Capsule endoscopy
- ☐ B) Left hemicolectomy
- ☐ C) Repeat angiography
- ☐ D) Repeat radiolabeled erythrocyte scan
- ☐ E) Total abdominal colectomy



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27. A 62-year-old woman has had halitosis and progressive dysphagia with spontaneous regurgitation of undigested food over the past 2 years; she has lost 2.3 kg (5 lb) during this period. There are no lesions in the mouth. Abdominal examination shows no tenderness or masses. Test of the stool for occult blood is negative. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) 24-Hour esophageal pH monitoring
- ☐ B) Esophageal manometry
- ☐ C) Barium swallow
- ☐ D) Ultrasonography of the abdomen
- ☐ E) Esophagogastroduodenoscopy



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28. A 57-year-old man comes to the physician because of a 4-week history of mild epigastric pain and nausea 1 hour after meals. He has not had vomiting, heartburn, a bitter taste in his mouth, weight loss, or change in the color of his stools. He has no history of serious illness and takes no medications. He drinks coffee with dinner. He does not smoke. Examination shows no abnormalities. Test of the stool for occult blood is positive. Which of the following is the most appropriate next step in management?

- ☐ A) Recommend avoidance of caffeine
- ☐ B) Serum *Helicobacter pylori* antibody assay
- ☐ C) Upper gastrointestinal series
- ☐ D) CT scan of the abdomen
- ☐ E) Esophagogastroduodenoscopy



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29. A 53-year-old woman comes to the emergency department because of increasingly severe abdominal cramping and malaise. She has a 3-year history of ulcerative pancolitis, treated with sulfasalazine. Four days ago she had hematochezia and an increase in the frequency and volume of bowel movements; treatment with prednisone resulted in no significant improvement. Her temperature is 37.6°C (99.6°F), pulse is 110/min, respirations are 24/min, and blood pressure is 110/70 mm Hg. Abdominal examination shows moderate distention, diminished bowel sounds, and diffuse tenderness without rigidity on palpation. Supine and erect x-rays of the abdomen show gas distending the colon. Which of the following is the most likely diagnosis?

- ☐ A) Carcinoma of the colon with obstruction
- ☐ B) Dysplasia of the colon with hemorrhage
- ☐ C) Enterocolonic fistula
- ☐ D) Gastroenteritis
- ☐ E) Toxic megacolon



30. A 72-year-old man who is in the hospital has had abdominal cramps and bright red rectal bleeding for the past 6 hours. During this period, he has passed three bloody, moderately sized stools. Two days ago, he underwent surgical resection of an infrarenal abdominal aortic aneurysm. He had a hemorrhage during dissection of the aneurysm and had hypotension for the first 20 minutes of the operation. He received 6 units of packed red blood cells. His initial postoperative course was uncomplicated, and he was extubated within 24 hours. Current medications are intravenous cefazolin and subcutaneous heparin. His current temperature is 38.2°C (100.8°F), pulse is 100/min, and blood pressure is 124/80 mm Hg. Abdominal examination shows mild distention and moderate left lower quadrant tenderness. Which of the following is the most likely diagnosis?

- ☐ A) Aortoenteric fistula
- ☐ B) Cecal angiodysplasia
- ☐ C) Diverticulosis
- ☐ D) Ischemic colitis
- ☐ E) Pseudomembranous colitis



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31. A 77-year-old woman is brought to the emergency department because of abdominal distention and obstipation for 2 days. She has no history of an abdominal operation. She takes 2 mg of diazepam three times daily. Test of the stool for occult blood is negative. An x-ray of the abdomen is shown. Which of the following is the most likely diagnosis?

- ☐ A) Carcinoma of the colon
- ☐ B) Diverticulitis of the colon
- ☐ C) Ileocolic intussusception
- ☐ D) Ischemic colitis
- ☐ E) Volvulus of the sigmoid colon





32. A 37-year-old woman comes to the physician because of a 6-month history of moderate rectal pain, rectal itching, and bleeding with bowel movements. She is otherwise healthy and takes no medications. On digital rectal examination, the rectal tissue is tender and inflamed. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Barium enema
- ☐ B) Endorectal ultrasonography
- ☐ C) CT scan of the abdomen
- ☐ D) Anoscopy
- ☐ E) Colonoscopy



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33. A 42-year-old woman comes to the emergency department because of increasingly severe abdominal pain for the past 4 hours and a temperature of 39°C (102.2°F) and shaking chills for the past 2 hours. The pain began 1 hour after eating dinner. During the past 2 weeks, she has had 30-minute episodes of epigastric and right upper quadrant abdominal pain after eating. Treatment with antacids sometimes relieves the pain. She now is lying still on the examination table. Her temperature is 39.5°C (103.1°F), pulse is 110/min, respirations are 22/min, and blood pressure is 90/60 mm Hg. Examination shows scleral icterus and jaundice. Abdominal examination shows right upper quadrant tenderness to percussion and involuntary guarding; bowel sounds are normal. Urinalysis is positive for bile. Which of the following is the most likely diagnosis?

- ☐ A) Cholangitis
- ☐ B) Cholecystitis
- ☐ C) Gastritis
- ☐ D) Hepatitis
- ☐ E) Pancreatitis



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34. A 57-year-old woman comes to the emergency department because of a 1-hour history of increasingly severe left-sided chest pain that is worse when she inhales. Six hours ago, she underwent upper gastrointestinal endoscopy because of a 1-month history of progressive dysphagia. Results showed a benign-appearing stricture of the distal esophagus that was balloon-dilated. Her only medication is esomeprazole for gastroesophageal reflux disease. She does not smoke cigarettes or drink alcohol. She appears anxious. Her temperature is 37.9°C (100.3°F), pulse is 110/min, respirations are 20/min, and blood pressure is 110/65 mm Hg. Examination shows splinting of the left chest during inspiration. Cardiac examination shows a regular tachycardia. Abdominal examination shows no abnormalities. Chest x-ray shows a small left pleural effusion. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Water-soluble contrast esophagography
- ☐ B) CT scan of the chest
- ☐ C) Repeat upper gastrointestinal endoscopy
- ☐ D) Thoracentesis
- ☐ E) Thoracostomy



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35. A 25-year-old man has postoperative pain 6 hours after a mesh repair of an inguinal hernia under local (lidocaine) anesthesia. Which of the following is the most appropriate analgesic pharmacotherapy for the incisional pain?
- ☐ A) Oral aspirin
 - ☐ B) Oral diazepam
 - ☐ C) Oral morphine sulfate
 - ☐ D) Oral oxycodone-acetaminophen compound
 - ☐ E) Transcutaneous fentanyl
 - ☐ F) Infiltration of the wound with bupivacaine

36. A 5-week-old boy is brought to the physician because of a 4-week history of progressive jaundice. He was born at term following an uncomplicated pregnancy and delivery. His mother says that his stools are white. He is at the 50th percentiles for height and weight. His temperature is 37.4°C (99.3°F). Examination shows moderate jaundice. The liver edge is palpated 4 cm below the right costal margin and is firm and slightly nodular. Serum studies show:

Bilirubin, total	12.4 mg/dL
Direct	8.3 mg/dL
Alkaline phosphatase	551 U/L
AST	51 U/L
ALT	48 U/L
α ₁ -Antitrypsin	normal

A TORCH assay is negative. An ultrasonography of the abdomen does not show a gallbladder. A technetium 99m scan shows prompt uptake throughout the liver and no gallbladder; no tracer appears in the bowel. Which of the following is the most likely mechanism of this patient's jaundice?

- ☐ A) Decreased bilirubin conjugation by hepatocytes
- ☐ B) Decreased bilirubin excretion by hepatocytes
- ☐ C) Decreased bilirubin uptake by hepatocytes
- ☐ D) Impaired bile flow through ducts
- ☐ E) Increased bilirubin production



37. Twenty-four hours after undergoing splenectomy for trauma sustained in a motor vehicle collision, a previously healthy 25-year-old woman has oliguria. A urinary catheter is in place, and her urine output has been 20 mL/h during the past 3 hours. Intraoperative complications included a 30-minute period of hypotension and a total blood loss of 2.5 L requiring 4 units of packed red blood cells. Currently, her temperature is 38.0°C (100.4°F), pulse is 100/min, respirations are 14/min, and blood pressure is 120/80 mm Hg. The lungs are clear to auscultation, and breath sounds are heard bilaterally. Abdominal examination shows no distention. Bowel sounds are absent. Laboratory studies show a hematocrit of 28%, serum urea nitrogen concentration of 30 mg/dL, and serum creatinine concentration of 2.5 mg/dL; her serum electrolyte concentrations are within the reference ranges. Central venous pressure is 8 cm H₂O (N=5–8). Which of the following is the most appropriate next step to confirm the cause of the oliguria?

- ☐ A) Administration of spironolactone
- ☐ B) Measurement of plasma free hemoglobin concentration
- ☐ C) Measurement of urine sodium and creatinine concentrations
- ☐ D) Renal ultrasonography
- ☐ E) Replacement of the urinary catheter



38. A 32-year-old man is brought to the emergency department 1 hour after sustaining extensive injuries in a motor vehicle collision. His pulse is 100/min, respirations are 15/min, and blood pressure is 120/70 mm Hg. Cardiopulmonary examination shows no abnormalities. The lower abdomen is tender. There is blood at the urethral meatus. On rectal examination, the prostate is not palpable, and there is no blood. Which of the following is the most appropriate next step in evaluation of the urinary tract?

- ☐ A) Cystography
- ☐ B) Intravenous pyelography
- ☐ C) Retrograde urethrography
- ☐ D) Ultrasonography of the pelvis
- ☐ E) CT scan of the pelvis
- ☐ F) Insertion of a urinary catheter



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39. A 32-year-old woman comes to the physician because of bloody discharge from the right nipple for 2 weeks. There are no breast masses or axillary or supraclavicular adenopathy. Mammography shows normal findings. The nipple and areola appear normal. Which of the following is the most likely diagnosis?
- ☐ A) Ductal carcinoma in situ
 - ☐ B) Fibroadenoma
 - ☐ C) Intraductal papilloma
 - ☐ D) Paget disease of the breast
 - ☐ E) Prolactin-producing pituitary adenoma



40. An otherwise healthy 40-year-old woman, gravida 2, para 2, comes to the office 2 weeks after she noticed a tender mass on her left breast while showering. Menarche was at the age of 12 years. The patient had her first child at the age of 28 years and underwent tubal ligation after the birth of her second child 8 years ago. The patient's maternal aunt was diagnosed with breast cancer at the age of 75 years. Examination of the patient shows no lymphadenopathy. The breasts appear normal and symmetric. Palpation of the breasts shows normal breast tissue bilaterally; there is a 2-cm firm mass on the upper outer quadrant of the left breast with no skin or chest wall attachment. Mammography confirms a solitary, well-defined mass in the same location. Ultrasonography of the breasts shows that the mass is fluid-filled and compressible with posterior shadowing. Which of the following is the most appropriate next step in management?

- ☐ A) Excisional biopsy
- ☐ B) Incisional biopsy
- ☐ C) Mammography-localized core needle biopsy
- ☐ D) MRI
- ☐ E) Needle aspiration



41. A previously healthy 42-year-old woman comes to the office because of a painful mass in the right breast for 3 weeks. Menarche was at the age of 12 years. She had her first child at the age of 22 years. Examination shows a 1-cm, firm, tender mass in the upper outer quadrant of the right breast; there is no lymphadenopathy. Examination of a biopsy specimen shows fibroadenoma and lobular carcinoma in situ. Over the next 10 years, this patient is at greatest risk for which of the following?

- ☐ A) Bone metastasis
- ☐ B) Fibrocystic breast condition
- ☐ C) Intraductal papilloma
- ☐ D) Invasive ductal carcinoma
- ☐ E) Ovarian metastasis



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42. A previously healthy 21-year-old man is brought to the emergency department 45 minutes after falling 12 feet from a barn loft. He landed straddling a wooden beam. His pulse is 120/min; other vital signs are within normal limits. Examination of the scrotum and penis shows edema and ecchymoses. Urinalysis shows numerous RBCs. Doppler ultrasonography of the testes shows a scrotal hematoma; there is no blood flow to the testes. An analgesic is administered. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Renal ultrasonography
- ☐ B) MRI of the pelvis
- ☐ C) Intravenous pyelography
- ☐ D) Cystoscopy
- ☐ E) Surgical exploration



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43. A 32-year-old man has a 6-month history of irritability, frequent bowel movements, and a 5.4-kg (12-lb) weight loss. Examination shows a single thyroid nodule. Serum thyroxine (T_4) concentration is 16.1 $\mu\text{g/dL}$, and serum thyroid-stimulating hormone concentration is less than 0.1 $\mu\text{U/mL}$. Which of the following is the most likely diagnosis?

- ☐ A) Defect in thyroxine (T_4) biosynthesis
- ☐ B) Graves disease
- ☐ C) Multinodular goiter
- ☐ D) Riedel thyroiditis
- ☐ E) Thyroid carcinoma
- ☐ F) Thyroid cyst
- ☐ G) Thyroiditis
- ☐ H) Toxic adenoma
- ☐ I) Triiodothyronine (T_3) thyrotoxicosis



44. Twenty-four hours after undergoing total thyroidectomy and lymph node dissection for metastatic papillary thyroid carcinoma, a 57-year-old woman has circumoral paresthesia and tingling of the fingertips. Examination shows hyperreflexia when the facial nerve is tapped. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Intravenous calcium gluconate
- ☐ B) Intravenous magnesium sulfate
- ☐ C) Oral calcium carbonate
- ☐ D) Oral magnesium oxide
- ☐ E) Oral vitamin D₃



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45. Two days after undergoing thoracotomy for a stab wound to the right hemithorax, a 25-year-old woman has oliguria despite adequate fluid replacement. During the operation, she received 4 units of packed red blood cells. Her hematocrit is 35%, serum urea nitrogen concentration is 12 mg/dL, and serum creatinine concentration is 0.9 mg/dL. Her oliguria is most likely caused by an increased concentration of which of the following hormones?

- ☐ A) ADH (vasopressin)
- ☐ B) Cortisol
- ☐ C) Epinephrine
- ☐ D) Glucagon
- ☐ E) Norepinephrine



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46. A 14-year-old girl has several generalized tonic-clonic seizures and a decreased urine output 24 hours after undergoing revision of an obstructed ventriculoperitoneal shunt. She is being mechanically ventilated, and she is receiving continuous morphine infusion for pain. She appears well hydrated. Her temperature is 37°C (98.6°F), pulse is 85/min, and blood pressure is 118/70 mm Hg; the ventilatory rate is 16/min. Laboratory studies show:

Serum	
Na ⁺	118 mEq/L
K ⁺	4 mEq/L
Urea nitrogen	3 mg/dL
Osmolality	248 mOsmol/kg
Urine	
Specific gravity	1.032
Sodium	192 mEq/L
Osmolality	800 mOsmol/kg
Urea nitrogen	600 mg/dL

Pulmonary artery catheterization shows a central venous pressure of 9 cm H₂O (N=5–8). Intravenous administration of which of the following is the most appropriate next step in management?

- ☐ A) Desmopressin
- ☐ B) Glucose
- ☐ C) Naloxone
- ☐ D) Pentobarbital
- ☐ E) 3% Sodium chloride



47. A 52-year-old man undergoes a transurethral resection of the prostate for urinary hesitancy and urgency. His postoperative diagnosis is benign prostatic hyperplasia. The operation was complicated by a 30-minute episode of hypotension that ultimately responded to treatment with fluids and vasopressors. Forty-eight hours after the operation, he is alert and in no acute distress. His temperature is 37.8°C (100°F), pulse is 80/min, respirations are 16/min, and blood pressure is 130/80 mm Hg. He is oriented to person, place, and time. Serum studies show a potassium concentration of 6.6 mEq/L, urea nitrogen concentration of 20 mg/dL, and creatinine concentration of 3 mg/dL. Preoperative laboratory studies showed normal findings. An ECG shows a normal sinus rhythm with a rate of 100/min, PR interval of 0.122 sec (N=0.12–0.20), QRS interval of 0.084 sec (N=0.10), and QT interval of 0.4 sec (N≤ 0.44). There are peaked T waves across the precordium. Which of the following is the most appropriate next step in management?

- ☐ A) Repeat measurement of serum potassium concentration
- ☐ B) Low-potassium diet
- ☐ C) Rectal sodium polystyrene sulfonate (Kayexalate) therapy
- ☐ D) Intravenous calcium chloride therapy
- ☐ E) Immediate hemodialysis



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48. Eight hours after laparoscopic cholecystectomy, an obese 55-year-old man has a pulse of 150/min and blood pressure of 60/40 mm Hg. Hematocrit is 28% (42% preoperatively). Serum creatine kinase activity is 200 U/L. MB isoenzyme of creatine kinase fraction is less than 2% (N<5% of total). Arterial PO_2 is 70 mm Hg. An ECG shows sinus tachycardia without other changes. Central venous pressure is 2 mm Hg. Which of the following is the most likely diagnosis?

- ☐ A) Addisonian crisis
- ☐ B) Myocardial infarction
- ☐ C) Postoperative bleeding
- ☐ D) Pulmonary embolus
- ☐ E) Septic shock



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49. A 45-year-old man is brought to the emergency department by paramedics 30 minutes after he was involved in a motor vehicle collision. He was the unrestrained driver. On arrival, he is oriented to person but not to place or time. He has no documentation in his wallet regarding his religious affiliation, but his friend says she thinks he might be a Jehovah's Witness. His family cannot be contacted. His temperature is 36.7°C (98°F), pulse is 110/min, respirations are 20/min, and blood pressure is 90/50 mm Hg. Fifteen minutes later, his pulse is 135/min, and blood pressure is 75/40 mm Hg. Abdominal examination shows distention and tenderness with rebound. Abdominal ultrasonography for trauma (FAST) shows blood in the hepatorenal recess. His hematocrit is 15%. Which of the following is the most appropriate next step in management?
- ☐ A) Immediate consultation with the hospital ethics committee
 - ☐ B) Intravenous administration of fluids only
 - ☐ C) Transfusion of packed red blood cells
 - ☐ D) Withholding treatment until the patient's Jehovah's Witness status can be confirmed



50. A 37-year-old woman remains ventilator-dependent 3 weeks after undergoing a gastric restrictive procedure. She has obstructive sleep apnea and restrictive lung disease. Preoperative arterial blood gas analysis on room air showed a pH of 7.35, P_{CO_2} of 65 mm Hg, and PO_2 of 45 mm Hg. She is currently receiving fentanyl and lorazepam for sedation to maintain acceptable arterial blood gas concentrations. Attempts to reduce the dosages of medication have resulted in life-threatening hypoxia and carbon dioxide retention. Tracheostomy is recommended to prevent permanent vocal cord damage from continued orotracheal intubation. Prior to the operation, no discussions took place concerning tracheostomy placement. She is unmarried, and her only living relative is a sister who resides in another state. Which of the following is the most appropriate course of action?

- ☐ A) Continue current ventilator support with orotracheal intubation
- ☐ B) Contact the patient's sister to discuss consent for the tracheostomy
- ☐ C) Contact the hospital ethics committee to give consent
- ☐ D) Discontinue sedation and determine if the patient can communicate her wishes
- ☐ E) Proceed with tracheostomy as a medical necessity